Lecture 13 - Inverses, LU decomp.

$AA^{-1} = A^{-1}A = I$

$Ax = b \iff x = A^{-1}b$

$A^{-1} = [x_1 \cdots x_n] \Rightarrow Ax_i = e_i$

Gauss-Jordan

$$\begin{array}{c} A \\ \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 4 & -6 & 0 & 0 & 1 & 0 \\ -2 & 7 & 2 & 0 & 0 & 1 \end{array}\right] \end{array} \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - 2R_1}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -8 & -2 & -2 & 1 & 0 \\ 0 & 8 & 3 & 1 & 0 & 1 \end{array}\right]$$

$$\begin{array}{c} U \\ \rightarrow \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -8 & -2 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right] \end{array}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 2 & -1 & -1 \\ 0 & -8 & 0 & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right] \begin{array}{l} R_2 \to R_2 + 2R_3 \\ R_1 \to R_1 - R_3 \end{array}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & \frac{12}{8} & -\frac{5}{8} & -\frac{6}{8} \\ 0 & -8 & 0 & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right] R_1 \to R_1 + \tfrac{1}{8}R_2$$

$$\rightarrow \begin{array}{c} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{12}{16} & \frac{5}{16} & -\frac{6}{16} \\ 0 & 1 & 0 & \frac{4}{8} & -\frac{3}{8} & -\frac{2}{8} \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right] \begin{array}{l} R_1 \to \frac{1}{2}R_1 \\ R_2 \to -\frac{1}{8}R_2 \end{array} \\ \quad I \qquad\qquad\qquad A^{-1} \end{array}$$

What happened:

$(EFG \cdots)A = I$

elimination matrices

$\Rightarrow EFG \cdots = A^{-1}$

$(EFG \cdots)[A \mid I]$

$= A^{-1}[A \mid I] = [I \mid A^{-1}]$.

Elimination with matrices

$A \to U$ steps:

1) $R_2 \to R_2 - 2R_1$: mult by E
2) $R_3 \to R_3 + R_1$: mult by F
3) $R_3 \to R_3 + R_2$: " " G

$GFE \cdot A = U \Rightarrow A = (GFE)^{-1}U = \underbrace{E^{-1}F^{-1}G^{-1}}_{L} U$

$$\underset{E^{-1}}{\begin{bmatrix} 1 & & \\ 2 & 1 & \\ & & 1 \end{bmatrix}} \cdot \underset{F^{-1}}{\begin{bmatrix} 1 & & \\ & 1 & \\ -1 & & 1 \end{bmatrix}} \cdot \underset{G^{-1}}{\begin{bmatrix} 1 & & \\ & 1 & \\ & -1 & 1 \end{bmatrix}}$$

$$= \begin{bmatrix} 1 & & \\ 2 & 1 & \\ -1 & -1 & 1 \end{bmatrix} = L$$

Elimination steps.

Another example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix} \longrightarrow U = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & \\ 3 & 1 \end{bmatrix} \qquad \boxed{A = LU}$$

To solve $Ax = b$, solve $\begin{cases} Lc = b \\ Ux = c \end{cases}$

Then: $Ax = L(Ux) = Lc = b$.

Triangular/echelon form

EASY.